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Energy-Based VAD vs. Neural VAD (Siler)

Energy-Based VAD

Energy-based VAD measures the root-mean-square (RMS) energy of each audio frame (typically 10–30 ms) and compares it against an adaptive noise floor. Frames above threshold are labeled speech; frames below are silence. [callmissed](#)

Limitations:

- Fails in noisy environments (TV, café, keyboard clicks) where background energy exceeds speech energy
- Cannot distinguish human speech from other loud sounds (door slams, music)
- High false-positive rate in telephony (G.711 μ -law narrowband) due to codec artifacts [proceedings.iclr](#)

Neural VAD (Sileró)

Neural VAD (e.g., Silero VAD v5) replaces hand-crafted energy thresholds with a recurrent neural network trained on thousands of hours of labeled speech across languages, accents, and noise conditions. The model outputs a per-frame speech probability $P(\text{speech}) \in [0, 1]$. [proceedings.iclr](#)

Advantages:

- Detects speech in <15 ms with high robustness to non-speech noise
- Learns vocal formant resonances (F1, F2, F3) and pitch harmonics (F0), rejecting sirens, dogs, typing [proceedings.iclr](#)
- Silero v5 achieves ~87.7% speech detection at 5% false-positive rate vs. ~63% for WebRTC GMM energy filter [aclanthology](#)

VAD TYPE	TECHNICAL BASIS	LATENCY	NOISE ROBUSTNESS	FALSE TRIGGER RATE
Energy-based (WebRTC legacy)	RMS energy + spectral flatness	<1 ms	Low	24.5% proceedings.iclr
Neural (Silero v5)	RNN on mel-spectrogram	1–5 ms	High	6.8% proceedings.iclr

Silence-Threshold Approach and the 500–700 ms Latency Tax

The **silence-threshold approach** waits for a fixed duration of continuous silence (e.g., 500 ms) before declaring end-of-turn. This is the default in most production STT APIs and voice pipelines. [autointerviewai](#)

Why 500–700 ms is a large fraction of the latency budget:

PIPELINE STAGE	TYPICAL LATENCY (P50)
VAD end-of-speech hold	300–700 ms (configurable) dev
ASR streaming first partial	100–300 ms hamming
LLM time-to-first-token	200–600 ms

PIPELINE STAGE	TYPICAL LATENCY (P50)
TTS time-to-first-audio	75–300 ms
Total	675–1,900 ms

At 500 ms silence threshold, VAD endpointing alone consumes **30–50% of a sub-1-second latency budget**, leaving insufficient margin for ASR, LLM, and TTS. [autointerviewai](#)

The 200 ms human baseline: In natural human conversation, the average gap between turns is ~200 ms (Stivers et al., PNAS 2009). A 500 ms silence threshold is **2.5× the human baseline**, making agents feel sluggish even if downstream stages are fast. [gradium](#)

Semantic Turn Detection

Semantic turn detection (semantic VAD, semantic endpointing) predicts end-of-turn from linguistic content rather than silence duration alone. It answers: *"Has the speaker finished a complete thought?"* instead of *"Has there been N ms of silence?"* [gradium](#)

How It Works

Semantic VAD analyzes:

- **Syntactic completeness:** Does the partial transcript end mid-clause ("I want to...") or at a grammatical boundary ("I want to book a flight.")? [gradium](#)
- **Intonation contours:** Falling pitch (F0 downdrift) signals completion; level/rising pitch signals continuation [proceedings.iclr](#)
- **Filler words and hesitations:** "um", "uh", "let me think" extend the pause threshold dynamically [gradium](#)
- **Question-answer adjacency pairs:** A question ("What's my balance?") implies an expected response, triggering faster endpointing [gradium](#)

Multi-Horizon Inactivity Prediction

Modern semantic VAD (e.g., Gradium STT) emits a **multi-horizon inactivity forecast** every 80 ms: an array of probabilities that the user will remain silent over the next 0.5, 1, 2, or 3

seconds. The agent reads whichever horizon matches its reactivity needs. [gradium](#)

```
vad: [0.12, 0.35, 0.68, 0.89] # P(inactive over 0.5s, 1s, 2s, 3s)
```

This allows:

- **Snappy assistants:** Read `vad [gradium]` (<https://gradium.ai/content/semantic-vad-voice-agents-turn-detection-2026>) (0.5s horizon) at threshold 0.30 → commit after 1–2 high-confidence frames [gradium](#)
- **Noisy telephony:** Read `vad [autointerviewai]` (<https://www.autointerviewai.com/blog/what-is-turn-taking-vad-in-voice-ai-interruptions-2026>) (3s horizon) at threshold 0.60 + require 3 consecutive high frames → robust to codec artifacts [gradium](#)

Latency Impact

Semantic VAD reduces endpointing latency from 500 ms to **100–200 ms** without increasing interruption rate, because it distinguishes hesitation pauses from true turn endings. [gradium](#)

Integrated ASR-Based End-of-Turn Signals

Some systems integrate turn detection directly into the ASR model. The ASR emits an **end-of-turn (EOT) flag** alongside partial transcripts, trained jointly on transcription and turn-completion prediction. [gradium](#)

Advantages:

- Zero additional model in the turn-detection loop: EOT signal comes from the same forward pass as the transcript [gradium](#)
- Context-aware: ASR has already seen the full acoustic context, so EOT is more accurate than a separate VAD [gradium](#)

Example: Gradium STT integrates semantic VAD natively, emitting `vad` arrays in the same WebSocket stream as transcripts. No separate endpointing component is needed. [gradium](#)

The Fundamental Trade-Off: Cutting Off vs. Making Them Wait

Turn detection faces an **unavoidable precision-recall trade-off**:

SETTING	EFFECT
Low silence threshold (200–300 ms)	Fast responses, but high interruption rate (cuts users off mid-thought) dev
High silence threshold (800–1,200 ms)	Fewer interruptions, but every turn carries 500+ ms of dead air → agent feels slow gradium

Why no single threshold works: A 400 ms pause mid-sentence ("I'd like to book a flight to... uh... Lisbon") is acoustically identical to a 400 ms pause at sentence end. Silence duration alone cannot distinguish hesitation from completion. [gradium](#)

Production Tuning Strategies

USE CASE	SILENCE THRESHOLD	MIN SPEECH DURATION	CONSECUTIVE HIGH FRAMES
Snappy assistant, clean audio	200–300 ms	500 ms	1
Default conversational agent	400–500 ms	800 ms	1–2
Phone IVR, noisy channels	600–800 ms	1,000 ms	3
Semantic VAD enabled	100–200 ms (effective)	N/A	1–2 high-confidence horizons

Dynamic tuning: Some systems adjust thresholds per user (fast talkers vs. slow thinkers) or per dialog state (open-ended questions → longer threshold; yes/no questions → shorter threshold). [gradium](#)

Backchannels ("mhm", "yeah") and Why They Must Not Trigger a Turn

Backchannels are short listener responses ("mhm", "yeah", "I see", "right") that signal active listening **without seizing the conversational floor**. In human conversation, backchannels occur ~2–4 times per minute and are essential for natural flow. [proceedings.iclr](#)

Why Backchannels Are Problematic for VAD

- **Acoustically identical to turns:** "mhm" is 200–400 ms of speech, indistinguishable from a short turn ("yes") to energy-based VAD [proceedings.iclr](#)
- **If misclassified as turn:** Agent stops speaking, waits for user response → awkward pause, or responds prematurely to a non-question [proceedings.iclr](#)

How Production Systems Handle Backchannels

1. **Duration filtering:** Ignore speech segments <300 ms unless followed by silence >800 ms [proceedings.iclr](#)
2. **Lexical classification:** Train a classifier to detect backchannel words/phrases ("mhm", "uh-huh", "got it") and suppress turn detection [proceedings.iclr](#)
3. **Prosodic cues:** Backchannels have flat or rising pitch contour (not falling like turn-yielding statements) [proceedings.iclr](#)
4. **Semantic VAD:** Recognizes that "mhm" after agent's explanation is not a complete thought requiring response [gradium](#)

Example: User says "mhm" while agent explains a refund policy. Semantic VAD sees no syntactic completion, no falling pitch, no question structure → continues agent turn without pause. [gradium](#)

Benchmarks and Key Papers

SYSTEM	VAD ARCHITECTURE	SPEECH ONSET DETECTION	BARGE-IN SPEED	ACCIDENTAL
WebRTC Energy VAD	GMM energy filter	65 ms	280 ms	24.5% proceedings
Silero VAD v5	RNN	20 ms	85 ms	6.8% proceedings
OpenAI Realtime VAD	Server-side audio gating	40 ms	120 ms	4.2% proceedings
Deepgram Flux	Semantic endpointing	25 ms	75 ms	3.6% proceedings

Key references:

- **Stivers et al. (2009, PNAS):** "Universals and cultural variation in turn-taking in conversation" — establishes 200 ms average human turn gap [gradium](#)
- **Gradium Semantic VAD (2026):** Multi-horizon inactivity prediction integrated into STT [gradium](#)
- **Silero VAD (2021–2026):** Open-source neural VAD with ONNX runtime [proceedings.iclr](#)
- **OpenAI Realtime API (2024–2026):** Semantic VAD mode with function calling [gradium](#)

Production systems in 2026 increasingly adopt **hybrid architectures**: acoustic VAD for input gating + semantic VAD for turn completion, achieving sub-500 ms end-to-end latency with human-parity interruption rates. [gradium](#)